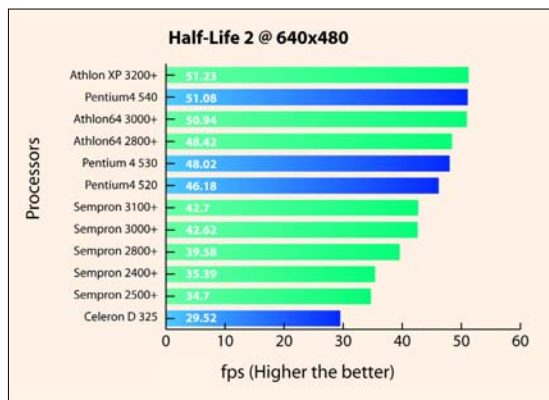




scores, we can infer that Half-life 2 scales well with your processor—the faster the CPU, the better the performance the game delivers.

And The Winner Is...

The Athlon64 3000+ makes it to the top thanks to its excellent performance and attractive pricing. Competing Intel processors lack that extra juice to take on the AMD Athlon64 3000+.

The Sempron range of processors provides the best value for money; at least in certain tests such as the DOOM 3 test, it gives even the high-end Intel processors a run for their money.

However, before you jump out to buy one of these, there are certain things to consider: the Socket 'A' versions of this processor, available as 2400+, 2500+, 2800+ and 3000+, are

Scoreboard					
Processors		Basic Specifications		Score (100)	Price *
	Speed	L2 Cache	FSB		
AMD Athlon64 3000+	2.01 GHz	512 KB	NA	65.23	6716
AMD Athlon64 2800+	2 GHz	513 KB	NA	61.99	6624
AMD Sempron 3100+	1.8 GHz	256KB	NA	60.95	5704
AMD Sempron 3000+	2 GHz	256KB	333 MHz	58.19	5566
AMD Sempron 2800+	2 GHz	256KB	333 MHz	58.90	4922
AMD Sempron 2500+	1.75 GHz	256KB	333 MHz	60.21	3404
AMD Sempron 2400+	1.67 GHz	256KB	333 MHz	63.30	2806
Intel Pentium4 540	3.2 GHz	1 MB	800 MHz	57.73	13900
Intel Pentium 4 530	3 GHz	1 MB	800 MHz	56.72	10900
Intel Pentium4 520	2.8 GHz	1 MB	800 MHz	55.29	9900
Intel Celeron D 325	2.53 GHz	256KB	533 MHz	50.91	4900

For the entire scoreboard, please visit www.thinkdigit.com

manufactured only to prolong the life of that platform. Therefore, consider buying them only if you are upgrading your older Athlon XP-based machine. If you are buying a new Sempron machine, we would recommend the new 3100+, based on the 754 socket, so you have the option of upgrading to Athlon 64 later.

While that was Sempron, the value processor from AMD, the Celeron D processors from Intel are a hopeless piece of silicon that you should not bet you hard-earned money on.

Choosing The Right Processor

As with any other element of your computer, define the typical usage that your processor will see before you choose one. If you are in academia, where your work involves scientific calculations, an AMD Athlon 64 processor would be ideal simply because of its raw number-crunching performance.

Similarly, people whose work involves 3D applications, image or video manipulation, that is, multimedia, should also go for an AMD processor. The high bandwidth between the CPU and memory gives them a good boost in these applications.

For media encoding or decoding tasks such as ripping movies to DivX, MP3 conversions, and the like, the Pentium 4 line, with its long pipelines, SSE 3 instructions and higher clock speeds are ideal.

If you are a gamer, we recommend the fastest AMD Athlon 64 processor available. Though the Pentium 4 3.2 GHz beats the Athlon 64 3000+ in the Half-Life 2 test, it does so simply because of the 1.2 GHz speed advantage the Pentium has over the Athlon 64 3000+. Moving to a slightly higher-

clocked Athlon 64 will paint a completely different picture.

MOTHERBOARDS

Two-and-half years ago, Intel, VIA and SiS were the only companies involved in the manufacturing of chipsets—the group of logic circuits that controls the functioning of a motherboard. Intel developed chipsets for their own processors, and their portfolio was perfectly tailored to suit the needs of their processors. On the other hand, solutions from VIA and SiS included chipsets for Intel as well as AMD processors, like the KM400, KT400A, SIS760 and others. nVidia and ATi have since joined this elite club of chipset manufacturers and now, with five companies offering chipsets, choosing the right motherboard is a challenge.

Amidst the transition the PC industry is going through, buying a motherboard based on older technology should be avoided at all costs. This section will shed light on how to tackle this transition. Here we help you choose the right motherboard that not only gives you the best performance but also provides enough headroom for upgrades.

The motherboards have first been divided on the basis of platform, namely Intel and AMD, and further classified according to chipset. Let us first uncover some facts about Intel chipsets.

Intel Motherboards

Motherboards based on Intel's platform have always been known for their stability, performance and reliability. This is to be attributed to the fact that Intel's chipsets are perfectly tailored to their processors. While some years ago, it was only Intel who manufactured chipsets for their own processors, VIA, SiS, and now nVidia have joined in to give con-

Price versus Performance



In this graph, we have plotted a processor's performance against the price at which it is offered. Price is along the X-axis, and processor performance on the Y-axis. The green dots represent AMD processors, and the blue dots indicate Intel processors.

As you can see from the graph, as the price increases, so does performance; however, the increase in performance is not linear. AMD's Athlon64 3000+ has the highest performance and it is offered at roughly Rs 7,000. Intel's 3.2 GHz comes in at second place, but is steeply priced at Rs 14,000. The difference in these prices is significant; however, there is a mere five per cent difference in performance. In other words, at 50 per cent of the price, the Athlon 64 3000+ offers better performance by a margin of five per cent over Intel's 3.2 GHz. Similar inferences can be drawn by comparing other pairs of processors.